

Dubnium

105 Db Named after Dubna, the Russian town where it was first made, element number 105 is radioactive. Dubnium is an artificial element and currently has no uses beyond scientific research.



Albert Ghiorso (right) with James Harris, a fellow scientist

Controversial discovery

Two teams of scientists, one from Russia and the other from the US, claimed to have made dubnium at different times. The team in Russia was led by Georgy Flerov in 1968, and the US team was led by Albert Ghiorso in 1970. It wasn't until 1993 that the Russian team was credited with the discovery, and dubnium was officially named.

Seaborgium

106 Sg Only a few atoms of this artificial element have ever been made. They were first created in 1974 at the Lawrence Berkeley National Laboratory, in California, by making californium atoms collide with oxygen atoms. Seaborgium is currently used only in scientific research.

Naming seaborgium

American chemist Glenn T. Seaborg worked with various teams during his career, creating many artificial elements. At the time, it was unusual to name an element after a living person, but he was honored in this way in 1993, before his death in 1999.

Cartoon of Seaborg



Bohrium

107 Bh Used only in scientific research, this artificial element was discovered in 1981 by a team of German scientists firing chromium and manganese atoms at bismuth and lead. This element was named bohrium in 1997, despite worries that it might sound too similar to that of element 5, boron.



Niels Bohr in his laboratory

Niels Bohr

Bohrium was named in honour of Danish physicist Niels Bohr, who is best known for his work on the structure of the atom and his insight into atomic theory. His famous "Bohr model" of the atom earned him the Nobel Prize in Physics in 1922.

Hassium

108 Hs This element was named after the German state of Hesse, which is home to the Centre for Heavy Ion Research—the institution where hassium was made in 1984. To create hassium, a team of scientists, led by the German physicist Peter Armbruster, crashed lead and iron atoms together. Like all of the super-heavy artificial elements, hassium's atoms are radioactive, meaning that they quickly decay to form other elements. Therefore, a pure sample has never been collected, but it's expected to be one of the densest elements.